Stock price prediction system

A Project Report

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology in CSE

By

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Declaration

I hereby declare that the project report entitled "Stock price prediction system" submitted by me for the partial fulfillment of the requirements for the degree of Bachelor of Technology in CSE to the School of Engineering and Applied Science (SEAS), Ahmedabad University, is an authentic record of my own original work.

I further declare that this work has not been submitted in full or in part to any other institution or university for the award of any degree or diploma.

Date: _____ Signature: _____

Place: _____ Name: meet siddhapara

Industry Mentor Certificate

This is to certify that the project entitled "Stock price prediction system" has been carried out by meet siddhapara (AU2240243) as part of their internship at Grownited during the academic year 2025-2026.

The student has successfully completed the assigned tasks and their performance was found to be satisfactory.

Industry Mentor:
[Mentor Name]
Grownited

Date: _____ Signature: _____

Faculty Mentor Certificate

This is to certify that the project entitled "Stock price prediction system" submitted by meet siddhapara (AU2240243) is a bonafide work carried out under my supervision and guidance in partial fulfillment of the requirements for the degree of Bachelor of Technology in CSE from the School of Engineering and Applied Science (SEAS), Ahmedabad University.

Faculty Mentor:
[Faculty Name]
School of Engineering and Applied Science (SEAS)
Ahmedabad University

Date: _____ Signature: _____

Abstract

This project report details the development of a comprehensive Stock Price Prediction System and Screener application, tailored for the National Stock Exchange (NSE). The system leverages advanced Machine Learning techniques, specifically Long Short-Term Memory (LSTM) neural networks, to forecast next-day stock prices based on extensive historical data. In addition to deep learning predictions, the platform serves as an all-in-one financial assistant for retail investors. It features robust capabilities including real-time fundamental information retrieval, interactive technical indicators (such as MACD, RSI, and Bollinger Bands), an automated stock screener for breakout signals, and candlestick pattern recognition. The project demonstrates the successful integration of a machine learning backend with an intuitive, interactive frontend built using Streamlit. Key outcomes include the creation of a responsive, user-friendly dashboard, an AI-powered trading coach for strategy simulation, and a scalable SQLite database for efficient data management. The system empowers investors with data-driven insights and sophisticated analytical tools to make informed trading decisions.

Table of Contents

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Chapter 1: Introduction

1.1 | Project Definition

The objective of this project is to build an interactive, data-driven web application that serves as an all-in-one financial assistant for retail investors, specifically focused on National Stock Exchange (NSE) stocks. It integrates fundamental data, technical indicators, pattern recognition, and a deep learning (LSTM) model for next-day price forecasting.

1.2 | Project Objectives

- Develop an automated stock screener to identify breakout and consolidation zones.
- Implement and fine-tune a Long Short-Term Memory (LSTM) network capable of accurately predicting short-term stock price movements.
- Build a user-friendly frontend dashboard (using Streamlit) with a modern, responsive UI.
- Provide a simulated investment environment (AI Trading Coach) to help users practice trading strategies and receive real-time feedback.

[Figure 1.1: High-Level System Architecture Diagram]

Figure 1.1 illustrates the flow of data from Yahoo Finance API, through the data processing and machine learning layers, and finally to the Streamlit UI.

Chapter 2: Literature Survey

2.1 | Related Work

Numerous studies have been conducted on stock market prediction utilizing various statistical and machine learning methodologies. Traditional approaches relied heavily on ARIMA and linear regression. More recent advancements leverage deep learning, with Recurrent Neural Networks (RNNs) and their variant, LSTMs, showing significant superiority in capturing the non-linear, temporal dependencies inherent in financial time-series data.

2.2 | Tools and Technologies

Category	Technology Used
Programming Language	Python
Data Manipulation	Pandas, NumPy, yfinance
Machine Learning	TensorFlow, Keras (LSTM), Scikit-Learn
Frontend Framework	Streamlit
Database	SQLite

Table 2.1: Tools and Technologies used in the project.

[Figure 2.1: Comparison of Traditional vs Deep Learning Approaches]

Chapter 3: Methodology

The methodology for building the Stock Price Prediction System consisted of several discrete phases:

3.1 | Data Collection and Preprocessing

Historical stock data for NSE listed companies was retrieved dynamically using the Yahoo Finance API. Data points included Open, High, Low, Close (OHLC), and Volume metrics covering a 10-year period. Preprocessing involved handling missing values, calculating technical indicators (like Moving Averages), and scaling the features using Min-Max scaling to ensure optimal neural network convergence.

3.2 | Model Architecture

An LSTM network was selected for the predictive core due to its effectiveness in sequence prediction. The architecture involves multiple LSTM layers with Dropout layers interspersed to mitigate overfitting. The model was trained on rolling 60-day windows to predict the subsequent day's closing price.

[Figure 3.1: LSTM Model Layer Architecture]

3.3 | Project Timeline (Gantt Chart representation)

Phase	Task Description	Duration	Completion %
Phase 1	Requirements & Setup	Week 1-2	100%
Phase 2	Data Retrieval & DB	Week 3-5	100%
Phase 3	Technical Analysis Module	Week 6-8	100%
Phase 4	Screener & Patterns	Week 9-11	100%
Phase 5	LSTM Model Training	Week 12-15	100%
Phase 6	UI Polish & Testing	Week 16-17	100%

Table 3.1: Project Timeline and Milestones.

Chapter 4: Results

4.1 | Project Outcomes

The complete web application successfully loads and evaluates over 1,700 NSE stocks. The LSTM forecasting model demonstrates a high degree of accuracy in capturing the overall trend of stock movements, though volatility spikes remain challenging. The screener efficiently filters stocks based on predefined breakout parameters within seconds.

[Figure 4.1: Forecasting Module Performance Graph]

4.2 | My Contributions to the Project

During the internship at Grownited, my primary contributions included developing the frontend dashboard, writing the data retrieval pipelines, integrating the technical analysis libraries, and designing the LSTM prediction module from scratch.

4.3 | Learning Outcomes

- Gained deep insights into financial markets and algorithmic trading indicators.
- Acquired hands-on experience in building, tuning, and deploying deep learning models using TensorFlow/Keras.
- Learned how to develop and structure full-stack data applications using Streamlit.

4.4 | Real World Applications

This application can be utilized by retail investors to conduct rapid, comprehensive fundamental and technical analysis, mitigating the need for expensive premium charting software. The predictive insights also serve as a secondary confirmation tool for trade execution.

Chapter 5: Conclusion

The Stock Price Prediction System bridges the gap between complex machine learning techniques and everyday retail investing. By providing an intuitive interface coupled with deep learning-powered insights, the platform successfully serves its purpose as a comprehensive financial assistant. Future iterations may include real-time intraday data processing, integration of sentiment analysis from financial news feeds, and expanding the model's forecasting horizon to predict multi-day trajectories with confidence intervals.

Chapter 6: Example Chapter

6.1 | Only for inspiration

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6.2 | General formatting

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6.3 | Tables and figures

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Bibliography

[1] Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural computation, 9(8), 1735-1780.

[2] Python Software Foundation. (2026). Python Language Reference. Available at http://www.python.org

[3] Abadi, M. et al. (2015). TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems.

Appendix

Appendix A title

```
model = Sequential()
model.add(LSTM(units=50, return_sequences=True, input_shape=(x_train.shape[1], 1)))
model.add(Dropout(0.2))
model.add(LSTM(units=50, return_sequences=False))
model.add(Dropout(0.2))
model.add(Dense(units=25))
model.add(Dense(units=1))
model.compile(optimizer='adam', loss='mean_squared_error')
```